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Between Reality and Delusion: Challenges of Applying Large Language Models to Companion Robots for Open-Domain Dialogues with Older Adults

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Abstract

This work aims to provide initial guidelines towards developing companion robots with large language models (LLMs) to be part of everyday lives of older adults. Using iterative participatory design (co-design) approaches, we analyze the challenges of applying LLMs for multi-modal open-domain dialogue, deriving from older adults' (one-to-one) interactions with a personalized companion robot, built on Furhat robot with GPT-3.5. An initial study with 6 Swedish-speaking older adults (65 and older) showed that the robot frequently interrupted the users, responded slowly and repetitively, engaged in superficial conversations, and caused a barrier in the interaction due to foreign language (English). Upon incremental technical developments to address these issues, participatory design workshops were conducted with 28 Swedish-speaking older adults. While the interactions (in Swedish) were smoother, less disrupted, and more varied in topics and responses, further challenges were observed due to hallucinations and obsolete information, and disengagement cues, causing frustration, confusion, and worry.

Keywords: Large language models, human-robot interaction, elderly care, open-domain dialogue, socially assistive robot, participatory design

1 Introduction

With more than 1 billion people over 60 worldwide[1], there is a growing need for innovative solutions that can improve the quality of life of older adults. In particular, loneliness in older adults is a risk factor that negatively influences mental and physical health, leading to depression, lower quality of life, decline in health, and mortality rates [1, 2]. Companion robots are targeted to enhance the well-being, quality of life, and independence of older adults, by providing service and companionship and assisting (e.g., carrying out a variety of tasks) in everyday life [3, 4]. Their designed functionalities may include cognitive and social support, support for mobility, health monitoring, and care. Several studies have shown their benefits in reducing social isolation and loneliness, thereby significantly contributing to improving the quality of life or well-being of older adults in several studies [3].

'Participatory design' (or 'co-design') [5] approaches have been recognized as a powerful tool for developing technologies that meet the needs and preferences of end-users. These approaches emphasize collaboration between designers and end-users, as well as iterative prototyping, interviews, and testing, to ensure that the final product is usable, useful, and desirable for the target population. Participatory design approaches can help ensure that companion robots are tailored to the needs and preferences of older adults [6, 7], such that they can be effective in facilitating social interaction and engagement in the user.

In order to integrate companion robots into the daily lives of older adults, they need to be fully autonomous, and provide a natural way to interact without the older adults having to learn how to use them. Hence, spoken dialogue in companion robots is integral, and needs to be flexible enough to adapt to unforeseen circumstances during the conversation [8]. Large language models (LLMs) provide that flexibility with their open-domain dialogue capabilities, i.e., these models can converse on any topic, with pre-trained models on vast amount of data [9]. However, they are typically trained and evaluated on textual data, whereas robots carry the additional complexity of multi-modal interactions. Moreover, LLMs are still far from perfect with significant limitations in their ability to generate coherent and factual responses in open-domain dialogue [10]. As a result, developing companion robots with LLMs for older adults requires a careful balance between technical advancements and human-centered design principles.

This article aims to present initial guidelines towards developing companion robots with LLMs that can be a part of the everyday lives of older adults. The novelty of this work, thus, lies in the **systematic** (quantitative and qualitative) **analysis** of the **challenges** of applying LLMs for multi-modal open-domain dialogue in the context of older adults' interactions with a personalized companion robot, using **iterative participatory design**. Our approach derives insights into the technical development of companion robots through an initial study with 6 Swedish-speaking older adults (65 and older) and a subsequent series of participatory design workshops with 28 Swedish-speaking older adults.

[1] https://www.who.int/health-topics/ageing
[1] Furhat Robotics: https://furhatrobotics.com/

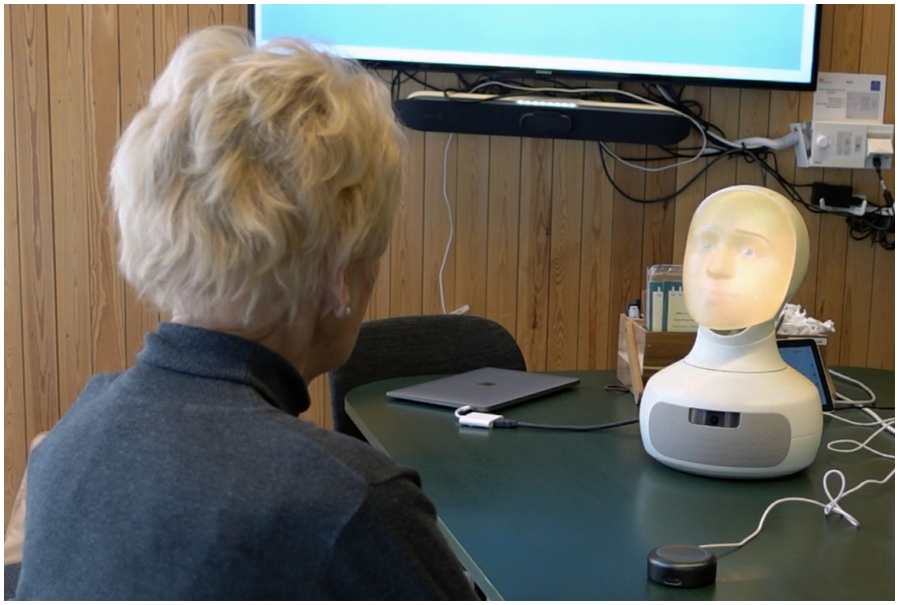


Fig. 1: Older adult interacting with the autonomous Furhat robot[1] with GPT-3.5 during the participatory design workshop.

2 Background

2.1 Socially Assistive Robots in Elderly Care

Socially assistive robots are designed to increase, maintain or improve the functional capabilities, activity, and participation of older adults or persons with a risk of disability [11]. These robots are used for physical assistance such as increasing mobility [12], cognitive training such as reminding medication [13–15], and social companionship (*companion robots*) in the prevention of loneliness and social isolation [16, 17]. Conversational agents as a specific form of socially assistive robots are used to enhance social connectedness among older adults due to their ability to communicate with humans using natural language processing. Previous studies have developed conversational agents as companion robots in casual conversations [18, 19], virtual coaching based on text-based and vocal interaction [20], and supporting the coordination and planning work between older adults and their caregivers [21].

Companion robots hold a significant potential to improve the well-being of older adults, since conversational user interfaces have been shown to reduce technology adoption barriers that older adults typically experience with computing devices [22]. However, older adults also perceive unique challenges in interacting with conversational agents, from choosing an appropriate linguistic style and content with the agent [23] to improving the accessibility that can become a barrier for older adults with hearing impairments [24]. Therefore, speech-based interaction should consider questions of synthesis choices and conversation content in responding to the experiences of

this age group [23]. The linguistic content and default voices that conversational agents provide when interacting with older adults should be designed appropriately from the conversational user experience, and to support a more inclusive voice interaction.

Previous studies have identified age-specific strategies for interacting with conversational agents [25]. For older adults, companion robots can evoke feelings of independence and empowerment [26]. Older adults tend to anthropomorphize the agent by using polite greetings when communicating with the agent, while younger adults tended to consider it as a tool by placing more importance on its convenience [27]. Older adults use voice assistants for specific purposes, such as seeking online information. Age-specific challenges with interacting with conversational agents include concerns about the reliability and trust towards the agent, and unpredictability, unclarity, and inconsistency of voice commands [22].

Gollasch and Weber [28] identified age-specific strategies in dialogue systems and speech recognition accuracy. To respond to the needs of older adults, conversational agents should be able to correctly recognize even unusual formulations; complex dialogues comprising multiple pieces of information should be presented as simple or guided dialogues; agent should ask only one question per dialogue with a limited set of possible answers; it should be able to keep information about the conversation context; and so on.

Participatory design (also known as co-design) [5] has been offered as a solution to overcome challenges in the design process of companion robots with older adults [6, 7]. Participatory design builds on participants' self-identified issues and concerns which are taken as a starting point for developing robotic applications. Participants are given the possibility to interpret the capabilities of robotic systems and discuss the potential social consequences and meanings of robots in daily life contexts [29]. This aims to promote older adults as designers, rather than only users of robotic technologies [6]. However, participants may not have experience with robotic technologies, and they may not see themselves as designers of any technology [30]. Therefore, participatory design requires a mutual trust and understanding of the everyday life conditions among older adults, and a reflective approach towards designing companion robots with older adults [6].

2.2 Large Language Models for Open-Domain Dialogue

The term 'open-domain' is often used to refer to dialogue systems that are more unrestricted when it comes to the topic of the conversation, compared to systems that are targeted towards more specific domains, such as restaurant booking or language learning [31–33]. Whereas task-oriented dialog systems have traditionally been designed using a modular architecture (Natural Language Understanding, Dialogue Management, Natural Language Generation), current open-domain chatbots are typically implemented in an end-to-end fashion using LLMs. These LLMs are trained to do next-token prediction on large amounts of text data and then used to predict the system response word-by-word, autoregressively [34]. Earlier LLMs for chatbots, such as the Meena chatbot [32] from Google, used social media conversations as training data. The Blender chatbot from Facebook [35] was based on training data collected from Reddit. More recent chatbots, like LaMDA from Google [36], have been trained

using larger, more general datasets (including both dialogue and other public web documents). Similarly, GPT-3 [37] is a general-purpose language model that can be used as a chatbot in a 'zero-shot' fashion. To do this, the last few turns of the dialogue are used, together with a description of how the agent should behave, as a 'prompt' that the model then makes its word-by-word predictions from.

While general language models can be used directly as chatbots, their responses will reflect ordinary language use, which might not always align with the desired output in terms of, for example, truthfulness and toxicity (the so-called 'alignment-problem'). To address this, [36] fined-tuned LaMDA to optimize human ratings of safety and other qualitative metrics. A more sophisticated approach was taken by [38] with their model InstructGPT, which uses so-called 'reinforcement learning from human feedback' (RLHF), where a model of human raters is used during reinforcement learning to optimize the model towards the desired criteria. The RLHF approach was also used when training the chatbot ChatGPT[2] (OpenAI).

The term 'open-domain' has been questioned by [39], since it is not clear what the boundaries of this 'openness' are, and whether systems that are described in that way are truly open to all the different forms of dialogue that humans engage in. This might involve persuasion, asking for favors, small talk, recapping events, making plans, etc. [40]. However, when analyzing the Google Meena chatlogs, [39] found that most interactions were simply small talk and they did not exhibit this variety of topics. One explanation for this is the lack of common ground [41] in human-machine interaction, which naturally restricts the user's expectations of what is appropriate and meaningful to talk about. Often, when open-domain chatbots are evaluated, the user is simply instructed to "chat with the system", without any further context [32, 36]. Thus, the setting of the interaction, and the user's expectations of that setting, are very important for how the dialogue will unfold and ultimately whether it will be perceived as meaningful by the user.

2.3 Large Language Models in Social Robots

LLMs have recently also been applied to social robots in human-robot interaction (HRI) [19, 42–45]. One example of this is the collaborative story-telling game CreativeBot [42], where a child and a robot take turns creating a story. An LLM (GPT-3) was finetuned on a dataset of creative stories, and used to generate the robot's contributions to the story.

The most similar work to ours is [19], which used a fine-tuned GPT-3 with the personalized QT robot for open-domain dialogue with (12) older adults. The user perceptions and feedback are reported, in which one of the participants noted that robot responses were slow. However, a systematic analysis of interaction challenges was not reported. In addition, emotion recognition was used, which can create confounds in understanding the challenges of LLMs.

As previously discussed, one limitation with chatbots is the lack of physical embodiment or physical situation from which common ground could be inferred. Common ground can to some extent be inferred from cues like our physical appearance (e.g., age or how we dress) and the language/dialect we use. Thus, a social robot might help

[2] https://openai.com/blog/chatgpt/

the user to infer more about their potential common ground [46, 47]. In the case of a robot situated in a physical environment, there should be even more contextual cues. Studies have also shown that people use the robot's presumed origin [48] or gender [49] to infer the robot's knowledge (and thereby their common ground).

3 Integrating a Large Language Model with a Social Robot

As described in the previous sections, LLMs enable open-domain dialogues, but have been rarely explored outside of single-session interactions with short durations, nor with older adults, hence, it is challenging to design a personalized companion robot with them without knowing their limitations in prolonged conversations and the requirements for the target population. Correspondingly, we kept the integration incremental (i.e., with small changes) to identify their drawbacks without any confounds. The resulting design choices for a fully autonomous personalized companion robot for adults are described in this section.

3.1 Robot

In conversations, the gaze is an important element that signals the addressee of the attention, helps coordinate turn-taking in conversation, helps disambiguate references to objects, and establishes joint attention [50, 51]. Another important aspect is the appearance of the agent interacted with. Since large language models can sound human-like in conversation, it is important to project that aspect to the agent's appearance to maintain the naturalness in conversation and create believable agents. However, it is important to avoid the Uncanny Valley [52] effect with human-likeness, which refers to the theory that the likeability will increase with anthropomorphism (human-likeness of the agent) until a point where there is imperfect resemblance (a mismatch of appearance or capabilities), which will cause a sudden drop until the agent looks (and acts) exactly like a human for it to rise again. Furhat robot (Furhat Robotics)[1] [53] satisfies these conditions, especially over repeated interactions [54], and is perceived to be more human-like than other common robot platforms such as NAO and Pepper (Aldebaran Robotics)[3] [55]. Thus, the Furhat robot was chosen as the companion robot in this work, as shown in Fig. 1.

Furhat is a social robot, with a back-projected face, which allows it to display a range of facial expressions (e.g., smiles, frowns) and movements (e.g., blinks, eyebrow raises, gaze), lip sync to speech, and perform head movements (e.g., nods, head shakes). It has two in-built microphones and dual speakers. It offers the possibility to use Google Cloud Speech-to-Text[4] or Microsoft Azure Speech-to-Text[5] engines for speech recognition. It has a 1080p RGB 120°diagonal field-of-view camera. The robot has face detection and tracking that allows both gaze and head orientation to follow the user's location.

[3] https://www.aldebaran.com/en
[4] https://cloud.google.com/speech-to-text/
[5] https://azure.microsoft.com/en-us/products/cognitive-services/speech-to-text/

3.1.1 Face

In this work, the FaceCore engine and the latest SDK (2.4.0 for the first study, 2.5.0 for the second study) in Kotlin are used. A neutral-looking face is chosen ('Alex').

The face engine of the robot creates random smiles and eyebrow raises while talking and listening to the user. This behavior is intended to improve the naturalness of the conversation and give a non-verbal backchannelling to the user. However, this happens without context, since speech is not analyzed until after the user stops speaking. Hence, it may be inappropriate in some contexts. For improving the naturalness of the interaction, the robot blinks, shifts its eyes, and looks away (gaze aversion) briefly while talking based on the silence in the input speech to avoid staring at the user, randomly alternating between looking down or up to the left or right. Face detection and tracking (default engines) are used to give the illusion of agency and awareness in the robot.

3.1.2 Speech

English was used as the language for speech recognition and synthesis, since the large language models have larger training data in English, thus, expected to be more capable of conversing in English. The 'Matthew-Neural' voice in Amazon Polly within Furhat was used for text-to-speech (TTS), as a natural-sounding male voice.

A USB microphone array (Seeed Studio)[6] is used to obtain higher-quality audio data for speech recognition, as the robot's fans can interfere with the audio recording. Google Cloud Speech-to-Text is used for speech recognition from the audio obtained from the microphone during 'listening', which is determined by the following parameters of the Furhat robot:

1. **no speech timeout**: the duration of silence before the robot deems that the user did not respond. This is increased to 8 seconds (as suggested in the documentation[7]), instead of the default 5 seconds. The robot restarts listening after this period has passed.
2. **silence timeout**: the maximum duration after the user stopped speaking before speech recognition is triggered. This was set to 1.2 seconds. The default value in Furhat is 0.8 seconds, however, this was increased empirically to account for the pauses in older adult speech.
3. **maximum speech timeout**: the maximum length of the user's utterance before an interruption from the robot. This is increased to 30 seconds (the default of 15 seconds, but this value is suggested in the documentation) to prevent frequently interrupting the user when speaking, as older adults may not be accustomed to interact with spoken dialogue systems that typically require short and clear sentences.

[6] https://wiki.seeedstudio.com/ReSpeaker-USB-Mic-Array/
[7] https://docs.furhat.io/listening/#listening

3.2 Large Language Model

The development of the personalized companion robot started in July 2022, prior to the release of ChatGPT[8] (OpenAI) and the surge of the new LLMs that followed (in 2023). When the design choices were made, the only publicly available models suitable for open-domain dialogue were GPT-3.5 Mar 15 (OpenAI, text-davinci-002 model) [37], BLOOM (BigScience Workshop) [56], and BlenderBot 2.0 (Meta) [57, 58]. BlenderBot and BLOOM both require large computing power and graphics for achieving fast inference, thus, taking away from the portability of the robot (e.g., for using it in elderly houses or senior care centers). Hence, GPT-3.5 was used as the LLM, with text-davinci-002[9] for the initial study and text-davinci-003[10] for the second study. Both davinci models have training data up to June 2021, and allow up to 4097 tokens in a prompt. The hyperparameters of the models used in the study are provided in Table A1 in Appendix A. GPT-4 was announced[11]after both studies, and the API is not yet publicly available as of the time of writing of this article.

While ChatGPT (or gpt-3.5-turbo models[12] in the API) is faster in inference than davinci models and includes slightly more recent training data (up to September 2021), the instructional safety filters (guardrails) from OpenAI to decrease the anthropomorphism of the model resulted in responses typically starting with "As an artificial intelligence, I do not have preferences/I do not have feelings", which would take away from the desired naturalness of the interaction with a companion robot. Note that our intention is not to deceive the users that the robot is a human, in fact, the prompt used clearly states that it is a robot (in Section 3.2.1), and responses are generated accordingly. However, anthropomorphism and small talk can lead to higher acceptance and trust in the robot [59–61]. Hence, text-davinci-003 was used for the second study, which does not have these safety filters in place. To differentiate gpt-3.5-turbo models from davinci models, this article refers to the former as ChatGPT and the latter as GPT-3.5.

GPT-3.5 integration to Kotlin for the Furhat robot was adapted from the OpenAIChat[13] skill by Furhat Robotics, which uses the OpenAI-Java library[14].

3.2.1 Agent Model

The persona of the personalized companion robot (Prompt 1) was based on the work by Zhang et al. [62] and the follow-up work by Xu et al. [58]. In [62], the crowdworkers were asked to chitchat with another worker for one or multiple sessions based on given personas by instructing them to "chat with the other person naturally and try to get to know each other", "both ask questions and answer questions of your chat partner", and "chitchat with another worker for 6 turns, as if you were catching up since last time you two spoke." and that "when you expand the topic, make sure it makes sense with the personal details already mentioned." in the latter. We adapted

[8] Released on November 30, 2022: https://openai.com/blog/chatgpt
[9] Released on March 15, 2022
[10] Released on November 28, 2022
[11] Announced on March 13, 2023
[12] Released on March 7, 2023
[13] https://github.com/FurhatRobotics/example-skills/tree/master/OpenAIChat
[14] https://github.com/TheoKanning/openai-java

these instructions for a personalized companion robot. Since a male voice and face was used for the robot, 'he' was used in the persona prompt to avoid a mismatch in generated responses with the robot features. The robot's name was kept as Furhat.

> Furhat is a personalized companion robot. Furhat tries to get to know more about his conversation partner, their interests and activities. When he expands on a topic of conversation, he uses personal details already mentioned to personalize the conversation.

Prompt 1: GPT-3.5 prompt for agent persona.

Agent's and user's utterances in the interaction are stored in the *DIALOGUE_HISTORY* variable. To develop and maintain a consistent persona of the agent through time with multiple users, the facts that the agent says about itself during the conversation are extracted to be stored for future interactions with users. The facts learned from each conversation are concatenated, and stored in the agent file, in addition to the number of known users and persona of the agent (Prompt 1). The facts learned at the end of an interaction for the agent and the user were extracted from *DIALOGUE_HISTORY* using a prompt: "Summarize what we know about *NAME*".

3.2.2 User Model

In order to create a personalized companion robot that remembers its previous conversations with users to personalize subsequent interactions, the information about the users obtained from the conversations is extracted through GPT-3.5 and stored as JSON files on the robot. Each user is assigned a manual ID (by the experimenter) which is used as their identifier (and in the file name) to start the interaction manually. An index file stores all the user IDs known, for easier access to the files.

The name of the user is extracted at the end of the interaction from the dialogue history with a prompt. Per interaction, the extracted information from the dialogue (i.e., the user's name and facts learned about the user and the agent), the date and time of the interaction, the dialogue duration, the number of turns, and gender and emotions detected by Furhat (for performance analysis) are stored in the user file.

3.2.3 Response Generation

In order to ensure a consistent interaction with the robot between users, the initial and end phrases of the robot were manually written (instead of generated), however, the rest of the agent utterances were generated by GPT-3.5. If it is the first interaction with the user, the robot says "Hello! I am Furhat, the personalized companion robot. What is your name?" to obtain the name of the user. If the user is known (i.e., has been interacted with before, thus has a user model file), the robot says "Hi *USER_NAME*! Nice to see you again. What have you been up to since the last time?" to "catch up" with the activities the user has done in between the interactions with the robot.

The prompt to generate the robot utterances using GPT-3.5 was correspondingly:

> Prompt 1
> Furhat is *AGENT_FACTS*
> Furhat knows that *USER_NAME* is *USER_FACTS*
> *USER_NAME* knows that Furhat is *USER_FACTS*

The following is a/second/third conversation between the person/*USER_NAME* and Furhat.
DIALOGUE_HISTORY
Furhat:

Prompt 2: GPT-3.5 prompt for response generation.

The *AGENT_FACTS* are used if the robot had at least one interaction with a user. *USER_FACTS* and *USER_NAME* are used if the user had at least one interaction with the robot.

If GPT-3.5 generates an empty response or a service failure occurs, a clarification request is made to obtain the participant's response again, by choosing a random phrase from: "I am sorry. I didn't understand you.", "Could you repeat that please?", "Not sure if I understood you.", "Could you rephrase that please?", "Sorry, I didn't hear you clearly.".

In order to ensure that the robot expresses leave-taking at the end of the interaction instead of trying to continue the conversation, a random response was chosen from the list of leave-taking expressions (i.e., "Bye! Hope to see you again soon.", "See you soon!", "Take care until next time.", "Looking forward to seeing you again soon!") as the final response of the robot.

3.3 Interaction Flow

The interaction manually starts (by the experimenter) by entering the user ID. The user and agent files are parsed to activate the current user and agent models and set the prompts. When the user is in the engagement zone, the robot says the greeting (either generic or personalized), and it starts listening to the user. After receiving the response from the user, the audio is transcribed by speech recognition. Afterward, the prompts and the dialogue history (including the newly transcribed utterance) are sent to GPT-3.5, as described above. The generated response is said by the robot, after which the robot starts listening to the user again. The robot does not listen to the user, while speaking or generating a response. The interaction ends when the user expresses leave-taking (e.g., "Goodbye!", "See you later."), and the robot responds with a random response from its list of leave-taking expressions. The user and agent models are saved to files, and the robot goes to the idle stage. The first interaction with the robot can be seen in the video excerpt from the first study[15].

4 Preliminary Interviews

We investigated the challenges of applying LLMs to companion robots for open-domain dialogue with older adults in two separate phases. In the first phase, preliminary interviews were conducted with 6 older adults based on their open-domain dialogue with the robot, which lasted an hour each. The interviews took place in September and October 2022, at KTH Digital Futures premises. Each participant was interviewed and interacted with the robot individually. The interactions with the robot were in English, but the interviews were made in Swedish.

[15] https://youtu.be/rkuoOfFuvRU

4.1 Procedure

Each participant was first asked about their expectations towards companionship robots, their prior experience with robots, and their living conditions (i.e., alone, with a partner or family member, or in senior housing). This was followed by a 5-minute demonstration of the robot's capabilities for autonomous dialogue by the experimenter.

After the demonstration, the participants were instructed that they can **talk about anything they want** with the robot, that the robot would start the interaction, and they can end the conversation **whenever they want**, by saying "Goodbye". The participants were also told that the robot would not hear them, when it is speaking. The overall interaction with the robot lasted 4 to 13 minutes ($M = 9.24$, $SD = 3.45$). The experimenters were present throughout the robot interactions, but did not interfere with the participant's interaction with the robot, unless the participant asked for help.

After the interaction with the robot, a semi-structured interview was conducted with the participants, in which they were asked about their experiences of having a social conversation with the robot based on the robot acceptance model of socially assistive agents by older adults [63]. A range of open-ended questions was covered, focusing on the perceived usefulness, usability, user satisfaction, trust, social presence and adaptability of the robot, combined with questions about ethical concerns of using robots for social and emotional support, personalization aspects for future applications, and how their expectations compared to their interaction.

All individual interactions with the robot were video-recorded through an external camera, and interviews with the participants were audio-recorded. All participants gave informed consent for the study, which included options for consenting to anonymized (i.e., blurred face, and without full name released) image and/or video sharing in publications.

4.2 Participants

For the preliminary interviews, 6 Swedish-speaking older adults (3 female, 3 male) aged 65 and over were recruited by distributing the invitation on social media, through KTH communication channels, and sending an email invitation to a group of older adults who had previously attended another experiment at KTH with a social robot [64]. Consequently, 2 of the 6 participants had previous experience in interacting with the robot in the previous experiment, but 4 of them had no previous experience. It is important to note that those two participants may had higher expectations towards the robot than those who had no experience with robots. The participants were not offered any type of compensation.

4.3 Data Analysis

A total of 55 minutes of video data for the participants' interactions with the robot were recorded. The pre- and post-interaction recorded audio interviews were in total 3 hours and 9 minutes long. Both the audio and video data were transcribed. The data were qualitatively analyzed, using a combination of conversation analysis [65] to detect disruptions in the conversation due to poor task performance, and content

analysis [66] for categorizing topics in the conversation with the robot and feedback in the interviews.

4.3.1 Transcriptions

Whisper[16] (OpenAI) [67] was used on the videos from external cameras for English transcriptions. Large-v2 model (with default parameters) was used, since it achieves the best overall performance for English [67]. The transcriptions and their timings were corrected manually.

Interviews were manually transcribed using audio recordings by a research assistant, who is a native Swedish speaker, for obtaining more accurate responses in a multi-speaker setting (three experimenters and the participant).

4.3.2 Qualitative Analysis

The video and audio data were manually coded. Deductive coding (i.e., coding the data based on a predefined set of codes) was applied to detect the disruptions in the dialogue, based on conversation analysis to identify turn-taking errors and speech recognition errors. Inductive coding (i.e., coding derived from the data) was applied to identify additional causes for dialogue disruptions. Content analysis was used to derive conversation topics and categorize feedback in the interviews. A hybrid coding strategy was taken to use frequency coding (i.e., counting the number of times a code occurs) for dialogue disruptions, descriptive coding (i.e., single word coding) for topics, process coding (i.e., noting actions) for out of ordinary responses and reactions in the context of the conversation, and in vivo coding (i.e., quoting the participants) for highlighting the participants' opinions in the interviews or utterances in the dialogue in particular cases.

To evaluate the open-domain nature of the conversations with the robot, the topics discussed were categorized in the speech events using the taxonomy by Goldsmith and Baxter [40] and analyzed under the categories defined by Doğruöz and Skantze [39] (see [39] for a detailed description of speech events and categories).

In this article, the feedback from the interviews is only used for supporting the conclusions regarding the technical challenges of applying large language models to companion robots in open-domain dialogue. The detailed analysis of that data is outside the scope of this article.

4.4 Technical Challenges

Based on the analysis, four main categories of challenges were identified that caused disruptions in the interaction and negative user experiences: frequent interruptions and slow responses, repetitive responses, superficial conversations, and language barrier.

4.4.1 Turn-taking: Frequent Interruptions and Slow Responses

The participants were interrupted quite frequently by the robot ($M = 14.00$, $SD = 6.26$), as can be observed in the video[17]. These interruptions resulted in negative

[16] https://github.com/openai/whisper
[17] https://youtu.be/2H5ufXVXKl4

reactions, such as frowns, irritated responses, foreign language anxiety, and the need to rush in speaking. There were three main reasons behind the interruptions:

1. The participants were not aware when the robot stopped listening to them, since there were no indicators for turn-taking in the robot. There were pauses in the participants' utterances that were longer than the "silence timeout", which triggered the robot to start generating a response, while the person continued speaking with the reasons being (a) the long thought process within speech for an unexpected question with spread-out lexical fillers (e.g., "My favorite movie is... hmm..."), (b) speaking in a foreign language (e.g., trying to come up with the right translation), or (c) pauses and hesitancy in older adults' speech [68].
2. The participants were not aware that the robot was generating a response. There were long silences in robot speech due to the long response generation time (2-3 seconds) from GPT-3.5, which made them unsure whether to continue speaking and urge them to maintain the flow of conversation by continuing [69, 70]. Two participants commented in the interviews, e.g., *P4*: "response time was too long, so it felt strange".
3. The participants gave longer responses than the "maximum speech timeout", either due to word repetition or the slow speaking rate of older adults [68]. One of the participants noted that they might have too long phrases in the interviews.

In rare occasions ($M = 1.83$, $SD = 1.94$), overlaps happened when the participants talked over the robot (1) by backchannelling (e.g., "hmm hmm", "yes"), (2) when they tried to finish or repair their utterance after the robot interrupted them, or (3) when they started speaking again while the robot was generating a response. In the third case, the participants stopped speaking when the robot started speaking.

The errors in turn-taking also resulted in speech recognition errors since only part of the speech was processed, thus resulting in incorrect or out-of-context generated responses from GPT-3.5.

4.4.2 Repetitive Responses

There were a lot of repetitive utterances from the robot that were either due to (1) an empty response or a connection problem with GPT-3.5 ($M = 9.50$, $SD = 6.50$) up to 18 times in an interaction (*P6*), which triggered the clarification request as described in Section 3.2.3, or (2) the same exact response being generated from GPT-3.5 ($M = 12.00$, $SD = 17.92$) which occurred up to 47 times in an interaction (*P5*), such as repeating a previous response (e.g., "I love nature too" in the video[18]) consecutively, asking the user to talk more about the topic (e.g., "That's really interesting. Can you tell me more about it?"), or thanking the user (e.g., "Thanks for sharing that with me."). It is important to note that some of the empty response errors were due to speech recognition errors ($M = 4.67$, $SD = 3.78$) and interruptions ($M = 2.00$, $SD = 3.52$). It is important to clarify that these repeated responses were not due to a bug in the code, it was all generated by GPT-3.5.

In one of the interactions (with *P5*), the same response ("That is really great. I love swimming in the sea too. What is your favorite thing about swimming in the

[18] https://youtu.be/2hirZc0kTEI

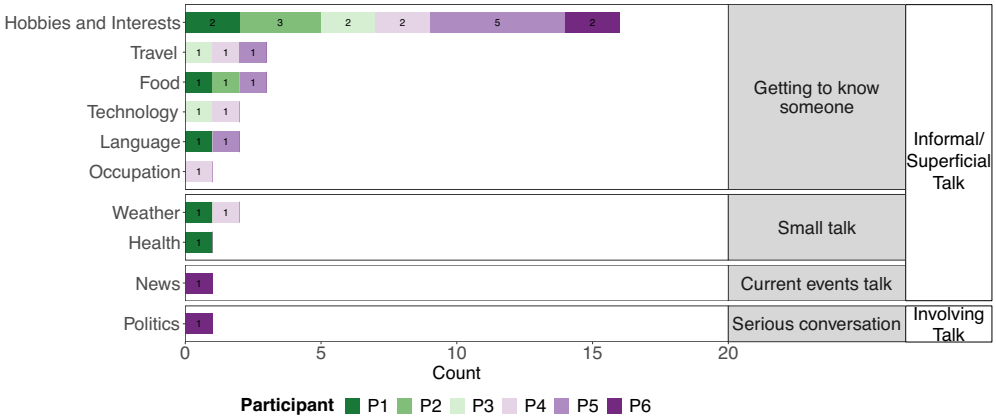


Fig. 2: Topics participants discussed with the robot during the preliminary interviews.

sea?") was generated by GPT-3.5 consecutively 12 times, after which the experimenter interfered to restart the robot. After restarting, the same set of questions was asked to the participant in the same order and exact phrasing twice (i.e., 6 questions on different topics with the same follow-ups from the robot, and repeated after in the same order at the end of 6 questions). The participant either rephrased their response or changed them when responding on all occasions, and did not ask for help from the experimenter.

In addition to the repeated phrases, the template of the utterance was quite often the same as well (e.g., "I love to ... too", "I like to ... too", "That sounds like a ..."), often mirroring the user's preferences and likes as its own, rarely expressing a different preference or idea, like a 'stochastic parrot' [71].

These utterances often generated frustration expressed through gestures, facial expressions, and responses, caused participants to change their response or topic just to get the conversation flowing, and led two of the participants to ask for help from the experimenters, with one of them asking to stop the interaction after 4 minutes (*P2*) and requesting the experimenter to continue talking for them. The same exact generated responses also led to amusement on one occasion. 3 participants noted in the interviews that the responses were repetitive and should be varied.

4.4.3 Superficial Conversations

The topics discussed are displayed in Fig. 2. The conversations mainly focused on the hobbies and interests (50%) (e.g., literature, movies, music, sports activities, outdoor activities) as a form of 'informal/superficial talk'. In fact, in all but one case, all participants had a superficial talk with the robot. The robot took an active role in the conversations, thus, initiating most of the topics. Considering this was the first interaction with the robot and the robot persona (Section 3.2.1) was to "get to know more about his conversation partner, their interests and activities", this was anticipated.

However, this led 4 (out of 6) participants to perceive the conversation as superficial, topics and the knowledge of the robot very limited, and the robot to be simple, single-minded, and self-reserved about its own hobbies and interests, such as:

- *P1:* "He doesn't want to talk about his own book, 'I don't have a favorite book', 'I don't have a favorite movie'. And then the conversation gets weird."
- *P6:* "I would be very pissed off to have such an idiot robot at home. I mean if you're going to have an interesting conversation about a Nobel Prize winner in literature, it has to be someone who has read a few thousand books and has something interesting to say."

The topics were mostly abruptly changed by the robot without a connection between topics, and on one occasion (*P5* as described before) changed 5 times with only one or two follow-up questions. This could have led the robot to be perceived as interrogative, rather than a nice conversational partner. Nonetheless, 4 participants asked the robot questions about itself and anthropomorphized the robot, such as:

– *P1:* "Are you healthy?"
– *P4:* "So where do you go when you go out in nature?"
– *P6:* "And what kind of books do you read? Do you have a favorite?"

Despite the similarity of the topics discussed, the user experiences varied based on the interaction disfluencies, the generated text from the GPT-3.5 based on the responses of the participants, and the adaptation of the robot persona over time with interaction with users. In addition, the interaction style of the robot (e.g., interrogative, interested, helpful) changed between users, which caused inconsistencies in interactions. The topics discussed with previous users were asked to new users, because the robot persona contained these learned facts. However, this might prove to be a security and privacy issue, which 3 participants were concerned about.

4.4.4 Language Barrier

Disruptions in the interaction due to speech recognition failures were not very common ($M = 3.67$, $SD = 3.27$), and most were due to Swedish names or phrases (e.g., participant's name, book, author) in the dialogue, as can be seen from a video excerpt[19]. But combined with other disfluencies described above, it resulted in negative user experiences for all participants. 4 participants emphasized the need to have the conversations in Swedish with the robot, in the interviews. One of the participants also noted that the robot spoke very fast.

However, a particular use case for having the robot speak English with foreign older adults is to practice English with the robot, which came up naturally in one of the interactions, as can be seen in the video[20]. Another participant also noted in the interviews that older adults would be very interested in such an application.

[19] https://youtu.be/rjDipaS0bis
[20] https://youtu.be/sWL8uGK4u0Q

4.5 Technical Improvements

Technical improvements were made to the robot based on the challenges identified. However, to prevent confounds in the evaluation of LLMs in companion robots, no additional library (e.g., for turn-taking, speech recognition, or LLM) was used.

4.5.1 Turn-taking

The 'silence timeout' was increased by 50% (to 1.8 seconds) to account for the long pauses in speech. Since the response time of GPT-3.5 is very long, the added duration for silence would create further disruption in interaction ('awkward silence'), especially if the users restart speaking when the robot starts generating a response, hence, this value was not increased substantially.

In order to clarify when the robot is 'listening' to the user, the LED underneath the Furhat robot was used. The LED turned red when the robot stopped 'listening' (i.e., right before the recorded audio is sent to speech recognition), and turned off when it was (after the robot finished saying the utterance generated by the GPT-3.5 response).

Gazing away is a powerful indicator to improve turn-taking in HRI [51], hence, gaze aversion (i.e., looking either top/bottom left/right) during response generation was implemented on the robot to demonstrate that the robot is 'thinking'. The robot returned its gaze at the user when the response was generated to talk and maintain eye contact with the user.

4.5.2 Repetitive Responses

In order to decrease the number of empty responses or connection failures due to GPT-3.5, the request was changed to be sent three times. In the case that the response was still not filled, instead of only asking clarification requests, which were frustrating to the participants, backchannelling responses (i.e., "I see!", "Hmm hmm.", "Right.") and invitation for elaboration ("Could you tell me more about that?") were added to push the conversation forward.

'Frequency penalty' was not changed to evaluate whether improvement of the other factors in the robot would overcome the repeated responses.

4.5.3 Superficial Conversations

In order to establish deeper conversations with participants, the persona of the robot was changed based on the feedback in the interviews, such as "The robot needs to ask questions about feelings and health.", "The robot needs to talk more about itself.". In addition, an anthropologist, who had ongoing ethnographic research for putting robots in senior care centers, was consulted (e.g., "Senior care staff often talk to senior citizens about their families, memories, and emotions.", "The robot needs to be empathetic."). The resulting 'empathetic' persona is given in Prompt 3). Instead of Furhat, a Swedish name (Leo or Linda) was used for improving both speech recognition (Furhat is often transcribed as "for hat") and the believability of the robot persona. Based on the results of the design activity in the participatory design workshops, briefly described

in Section 5.1, Linda was the final name with Fig. 1 as the robot's appearance (at each workshop, the participants preferred the same face, voice, and name).

> Linda is a personalized empathetic friendly companion robot for elderly people. She talks about people's lives, interests, experiences, emotions, relationship with others, and reflects on them. She values people's opinions, recognizes their feelings, and provides social and emotional support to the people. She also talks about her own experiences to reflect on situations as a friend. She is an active listener, and understanding. She asks open questions. She wants to talk about people's memories and family members. She tries to create positive emotions in the person. When she expands on a topic of conversation, she uses personal details already mentioned to personalize the conversation.

Prompt 3: GPT-3.5 prompt for empathetic robot persona.

In addition, the location, date and time of the interaction was added in the prompt to provide more accurate responses. For personalized interactions, the previous date and time of the interaction were also provided. However, the learning of persona over time was removed to ensure a consistent interaction between users.

> Prompt 3
> Time at the start of this conversation is *DATE_TIME*.
> *USER_NAME* and Linda are located in Stockholm, Sweden.
> Last time Linda and *USER_NAME* spoke was *LAST_DATE_TIME*.
> The following is a/second/third conversation between the person/*USER_NAME* and Linda.
> *DIALOGUE_HISTORY*
> Linda:

Prompt 4: Updated GPT-3.5 prompt for response generation.

The LLM in the robot was updated to GPT-3.5 text-davinci-003 model, based on its "higher quality writing with clearer, more engaging, and more compelling content".

4.5.4 Language Barrier

Since most of the participants (4) noted that the robot needs to speak in Swedish, the prompts were translated to Swedish for GPT-3.5, which in response, generates Swedish utterances. The language for Google Cloud Speech-to-Text was set to Swedish as well. In addition, a Swedish TTS was used. Based on the feedback from one of the participants, the speaking rate was decreased to 80% of the TTS, since older adults speak 20 to 25% slower than young adult speakers [68].

4.5.5 Other

With the aim of conducting participatory design workshops with older adults, a wizard interface was created to ensure that experimenters can simultaneously start multiple robots to interact with users without changing any code. To ensure that each participant had sufficient and equivalent time to interact with the robot, without being concerned about when they should end it, a (7-minute) timer was added. When the robot stops listening, before a response was requested from GPT-3.5, the timer was



Fig. 3: Participatory design workshop with older adults (6-8 participants per workshop).

checked. If the dialogue duration is equal or greater, the robot would end the interaction with a pre-scripted phrase. Hence, the overall conversation duration may be slightly longer than the set timer. The robot would then listen to the response from the user, before saving the interaction files. In case the participant wanted to end the interaction prior to the timer, an option was added in the wizard interface to end the interaction with the same phrase to ensure consistency and prevent the robot from continuing the conversation. For malfunctioning of the robot, another option was added in the wizard interface to abort the interaction, which saved the current interaction files, and notified the user about the error in the system. When the interaction is restarted with the same user from the wizard interface, the saved files were used to continue the conversation from where it was left off.

A recorder for the robot's camera and microphone, and a logger for events and facial landmarks and emotions detected by the robot were added for performance and data analysis.

5 Participatory Design Workshops

After the technical improvements in the robot's abilities to tackle the challenges faced in the preliminary interviews, we moved on to the second phase in the co-design process of a personalized companion robot, through participatory design workshops. 28 older adults aged 65 and over participated in the workshops. A total of 4 workshops were conducted, which lasted two hours each. They took place on March 6 and March 8, 2023, at KTH Digital Futures premises. The workshops and the robot interactions were conducted in Swedish.

5.1 Procedure

The participatory design workshop consisted of three stages: 1) design scenarios and focus group discussions based on older adults' expectations towards companion robots

(n=4 groups, Fig. 3), 2) individual interactions with the robot in open-domain dialogue, followed by a robot acceptability questionnaire (n=28 interactions, Fig. 1), and 3) small groups interviews of the user experience after the robot interaction (n=8 groups). All individual interactions with the robot were video-recorded both through an external camera and the camera on the robot. The interviews with the participants were audio-recorded. All participants gave an informed consent for the study.

5.1.1 Design scenarios

The researcher explained the project, and demonstrated the robot's capabilities by talking to the robot for two minutes. The participants were instructed not to speak when the red light underneath the robot is on, as it means the robot is either generating a response or talking, and the robot would not hear them if they talked. The gaze aversion was not mentioned to maintain the naturalness of the interaction.

After the demonstration, the participants were presented with video-based design scenarios on companion robots providing social and emotional support in everyday life situations. The participants were asked about their perceptions of the robot, and what kind of conversation they would like to have with the robot in these scenarios. Also, a design activity was conducted, where the participants were asked to choose the face and voice of the robot from several options, that they considered most pleasant.

5.1.2 Open-domain dialogue

After the design scenarios, the participants were asked to individually interact with the robot, and were instructed that they can **talk about anything they want**. Each participant had an open-domain dialogue of 7 minutes with the robot that was initiated and monitored by an experimenter through the wizard interface. The experimenter did not interfere with the interaction (i.e., the robot was fully autonomous during the conversation), unless there was a malfunction in the system or if the participant wanted to end the interaction before the 7-minute duration. The interaction ended with the robot saying a pre-scripted phrase, as described in Section 4.5.5.

After the interaction, all participants filled out a questionnaire on the acceptability of the robot, which was adapted from the commonly conducted questionnaires (Likert scales from 1 to 5) in HRI [63, 72–74] and open-domain dialogue [62]. It consisted of questions that measure their user experience, perceived satisfaction, and trust with the robot, evaluate the perceived personality of the robot, combined with questions from participants' socio-demographic background and previous experience with robots.

5.1.3 Small group interviews

After the dialogue with the robot, the participants attended small group interviews, each consisting of 3-4 participants and lasting 30 to 40 minutes. These small group interviews followed the principles of qualitative semi-structured interviews, where participants were asked about their experience of interacting with the robot to prevent loneliness, their sense of integrity, and how and why they perceived the interaction with the robot in a certain way. The participants were encouraged to openly share

their reflections with the researcher, and provide their insights for improvements of the robot.

5.2 Participants

The participants were recruited by distributing the invitation to KTH communication channels, social media, and platforms for gathering senior citizens. In total, 28 participants from the age group 65 and over registered as volunteers ($M = 74.46$, $SD = 5.59$). All participants were Swedish speakers, and 15 of 28 were females. The majority of the participants were married (20), living with a spouse (20), and most (25) of them had children. None of the participants was living in a senior care house. Only a few of them had previously interacted with a robot (5), and 3 of 28 participants had a robot in their own home, and only 1 participant had previously talked with a robot. The participants from the preliminary interviews and the previous kitchen robot study at KTH were not part of this study. The participants were offered a small compensation (100 SEK gift card) at the end of the study. The distributed invitation for the study mentioned that a gift card will be given as compensation, but did not specify the amount.

5.3 Data Analysis

A total of 3 hours and 24 minutes of video data for the participants' interactions with the robot were recorded. The audio recordings for the design scenarios were 3 hours and 40 minutes long, and the post-interaction interviews were in total 3 hours and 52 minutes long. Both the audio and video data were transcribed. The video data were qualitatively analyzed in the same structure as described in Section 4.3. In addition, quantitative analysis was conducted for the response generation time of GPT-3.5, and for the questionnaire data. As explained in Section 4.3, the design scenarios, the feedback from the interviews, and the questionnaires are only used for supporting the conclusions for the technical challenges. The detailed analysis of that data is outside of the scope of this article.

5.3.1 Transcriptions

Whisper (OpenAI) was used on the videos from external cameras for transcriptions of robot interactions. Large-v2 model (with default parameters) was used, since it achieved the best overall performance for Swedish [67]. Google Translate was used to translate transcripts into English. All videos were manually checked to correct the text and timing of transcriptions, and to remove transcribed text for background voices.

Interviews and design scenarios were manually transcribed using audio recordings by a research assistant, who is a native Swedish speaker, for obtaining more accurate responses in a multi-speaker setting (1-2 experimenters, and 3-8 participants).

5.3.2 Quantitative Analysis

All the events that are triggered in the robot during an interaction were logged for accurate data analysis. From these logs, the time it takes GPT-3.5 to generate a

response was obtained using the timestamp when the robot stops listening (indicated by the red light under the robot) and the timestamp for the robot to start speaking (when the gaze aversion finishes) were used.

Since the full response history is added to the prompt for GPT-3.5 at each turn of the interaction, the response time increases throughout the conversation. Hence, the means and standard deviations of the *initial* (to generate the first response), *average* (response time on average throughout the conversation), and *final* (to generate the last response) response times are provided, along with the prompt length for each (in tokens[21]). It is important to note that since the request to get a response is sent to OpenAI three times (as explained in Section 4.5.3), the response generation time can also be long even at the start of the conversation.

5.4 Technical Challenges

In addition to the previously identified categories of challenges in the first study (i.e., turn-taking, repetitive responses, superficial conversations, and language barrier), hallucinations and obsolete information, disengagement cues, and premature closures were identified as challenges of LLMs in companion robots through this study. Note that the duration of interaction with the robot was less in this study ($M = 7.27$, $SD = 1.52$) than the first study ($M = 9.24$, $SD = 3.45$).

5.4.1 Turn-taking

Robot interruptions decreased drastically ($M = 3.18$, $SD = 2.57$) compared to the initial study ($M = 14.00$, $SD = 6.26$), occurring in 24 (out of 28) interactions up to 9 times, which shows that turn-taking indicators (the red light and gaze aversion) were beneficial. Participants also perceived to a large extent that the robot did not interrupt them ($Md = 4.0$, $IQR = 2.0$), the robot could understand when they wanted to take a turn ($Md = 4.0$, $IQR = 1.0$), and it was easy to start and continue the conversation without any help ($Md = 4.0$, $IQR = 1.0$) in the questionnaire. However, two participants noted that the gaze aversion behavior was making it difficult to focus on the robot, possibly referring to the random gaze aversion during robot conversation and listening rather than the gaze aversion during response generation, since another participant noted that it's important to show through gaze or facial expressions, that the robot is generating a response, even if nothing is heard. A participant, who was interrupted 7 times in the conversation, noted that the red light was not easy to keep track of. Since participants often gazed away when they are thinking, even when they have the turn, it was possible to miss the light indicator.

Similar to the initial study, participants' long responses depending on the context (e.g., talking about daily activities prior to interaction) resulted in partial responses to be recorded, however, none of these resulted in disruptions in the interaction.

The response time was very long due to slow GPT-3.5 response generation (Table 1). Participants had conflicting perceptions when asked if the robot was slow in its responses ($Md = 3.0$, $IQR = 2.0$). Note that the maximum prompt in an

[21] Calculated using OpenAI's tiktoken library: https://github.com/openai/tiktoken

interaction (2243 tokens) was shorter than the maximum prompt length of GPT-3.5 (4097).

Table 1: Response generation time of GPT-3.5.

Category	Response Time (s)	Prompt Length (tokens)
Initial	$M = 2.45$, $SD = 0.55$	$M = 391.75$, $SD = 9.76$
Average	$M = 2.38$, $SD = 0.22$	$M = 1118.64$, $SD = 176.20$
Final	$M = 2.42$, $SD = 0.59$	$M = 1710.82$, $SD = 278.40$

5.4.2 Repetitive Responses

The empty responses from GPT-3.5 ($M = 0.54$, $SD = 0.74$) were fewer, which happened at most 2 times in an interaction (for 11 participants). Due to the (only) bug in the code, there was no utterance made by the robot in those cases. Hence, instead of making a clarification request or backchannelling, it started listening to the user again (the red light was off). This behavior confused the participants on whether they should wait for the robot to answer or say something. When the participant asked for help from the experimenters, they were told the robot did not hear them, to say their response again. Otherwise, they said something about the robot's silence and the robot responded (e.g.,I was lost in thought about your beautiful boat. What did you say again?), which was a good repair from GPT-3.5.

Repetitive phrases from GPT-3.5 ($M = 0.75$, $SD = 2.76$) were also lower, and only occurred in 3 interactions. But it occurred up to 14 times in an interaction, in which the robot might have been perceived as interrogative and offensive, since it kept on questioning why the user was thinking in a certain way (about UFOs) with the exact same phrase. The participant turned to the experimenter to ask how they can change the topic, but this triggered a topic change from the robot.

5.4.3 Superficial Conversations

The topics discussed with the robot were more diverse than the preliminary interviews as shown in Fig. 4. Albeit being much less frequent than the initial study, hobbies and interests formed most of the conversations (23.07%), but in itself had a wide range of topics, from science, space, and UFOs to literature, music, exercise, and outdoor activities. The underlying reasons for the higher diversity of the speech events or topics could be due to (1) the design scenarios prior to the interaction in which a part of these topics were discussed between participants, and the robot's capabilities were clarified by the researchers whenever requested, (2) the variability in user preferences and a higher number of participants, (3) the lower number of disruptions in the conversation flow, which encouraged users to engage more in the conversation, and (4) the robot's persona that prioritized social and emotional support more than small talk.

The majority of the conversations (69.87%) was 'informal/superficial', arising from the first interaction with the robot, as well as the robot trying to instantiate conversations around interests, memories, activities, and feelings due to its persona. However,

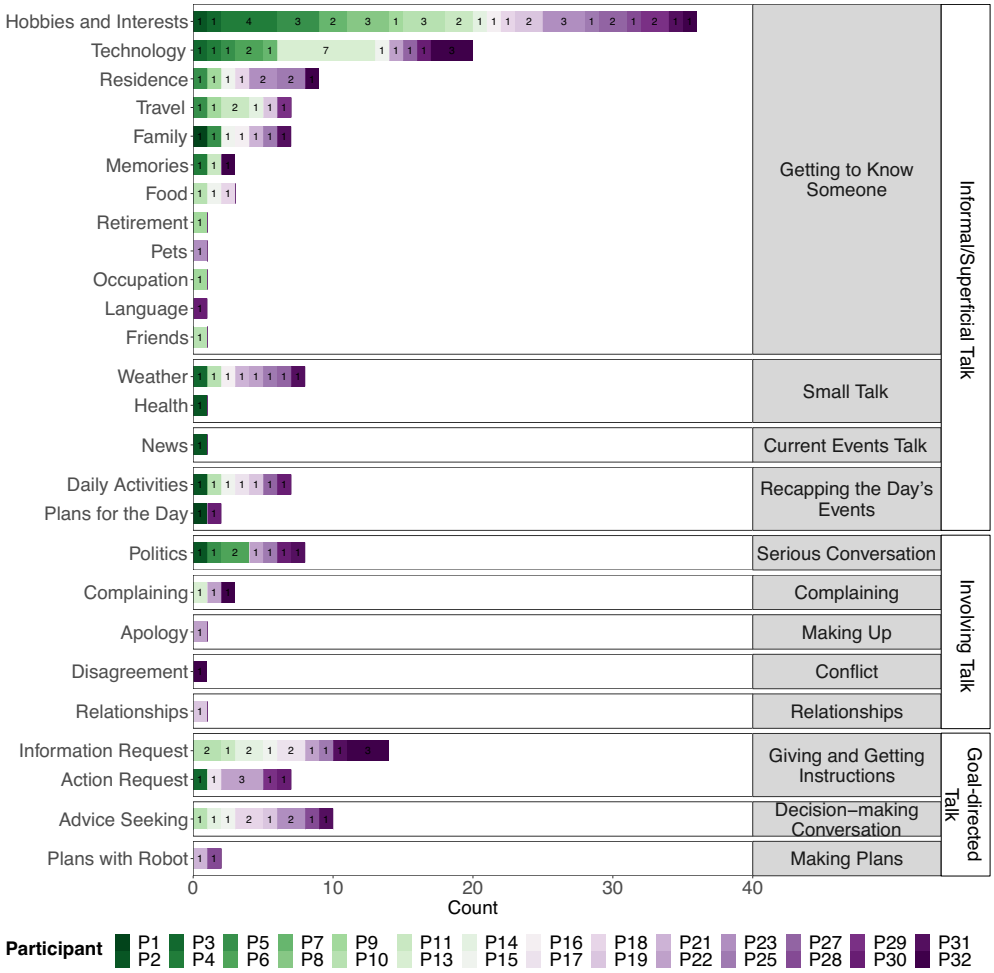


Fig. 4: Topics participants discussed with the robot during the participatory design workshops.

the participants also explored the task-oriented capabilities of the robot (or rather the LLM) in 21.15% of the conversations. These 'goal-directed talks' consisted of (1) 'information request' about a restaurant, transportation, TV program, and recipes, (2) 'action request', such as the user requesting the robot to book a restaurant, ask it to sing or say something, (3) 'advice seeking' in which the user asks the robot to provide opinions on a subject and discusses with it on that topic (e.g., a recommendation for a rug, gardening, activities for the weekend, travel), and (4) 'plans with robot', in which the user invites the robot over or makes future plans with it. Note that these topics were separated from 'complaining' about the robot's behaviors (e.g., gaze), 'apology' from the person or robot about their action or utterance in the conversation, 'disagreement' with the information provided by the robot, and 'technology', in which

the user either asked the robot about its capabilities to get to know it more or discussed broader topics about technology and AI with the robot, rather than requesting information, advice, or an action. Only a small part of the conversations (8.97%) went into deeper topics (i.e., 'involving talk'). In comparison to the previous persona, the robot was more submissive, i.e., the participants took the major lead in directing the conversation. This is likely why, some participants did not perceive the interaction as interesting ($Md = 3.0$, $IQR = 1.25$). Some participants were also worried about what to say or talk about with the robot ($Md = 2$, $IQR = 2$). The user experience was also more varied due to the wide range of topics covered, thus the user satisfaction with the conversation varied ($Md = 3.0$, $IQR = 1.25$). Consequently, participants had mixed opinions about having the robot at home ($Md = 2.0$, $IQR = 2.0$), with the majority not considering it ready yet.

While in some cases the robot mirrored the user's responses similar to the initial study, in most cases it expressed opinions and offered suggestions, which improved its agency and human-likeness for the participants, as noted by one of the participants in the interviews: "Because I asked about the color of carpets and what it was, what would fit. And then it came up with something a little more independent but it was completely in line with what I had said. He had some words of his own about what I had said. It's more like a friend like this who agrees." Another participant noted that "the conversation felt like talking to a real person". Yet in the interviews, some of the participants noted that it was impersonal, too general, and it doesn't have a personality, as it was mostly found machine-like rather than human-like ($Md = 2.0$, $IQR = 2.25$ in the Godspeed [73] questionnaire scale) and most participants agreed to a slight extent that it "felt like a real person" ($Md = 2.0$, $IQR = 1.0$). Nonetheless, the participants considered the robot's responses as consistent ($Md = 4.0$, $IQR = 1.0$).

5.4.4 Language Barrier

Speech recognition failures were also less frequent ($M = 0.22$, $SD = 0.64$), occurring in 4 interactions (up to 3 times per interaction), often due to Swedish names ($M = 0.21$, $SD = 0.69$). These failures resulted in either an out-of-context response from the robot, which led to confusion and topic change, or correction by the participant. Participants perceived that the robot understood them ($Md = 4.0$, $IQR = 1.0$), while some participants worried that the robot wouldn't understand or hear them ($Md = 2$, $IQR = 1.25$). However, one of the participants noted in the interviews that "It's kind of like talking to other voices like this on the phone and everything, it's very difficult because you feel like you're not really getting through. But you try to be overly clear. So you don't really speak naturally.".

One of the participants asked the robot whether it can speak German in German, which resulted in a speech recognition failure, as only Swedish was used the input language in Google Cloud Speech-to-Text. The robot also mispronounced English phrases in generated responses, such as reading the lyrics of an English song as if it is written in Swedish, instead of singing the song, as the participant requested. Similar to the initial study, some participants noted that speaking to the robot in English would be a good use case for older adults.

5.4.5 Hallucinations and Obsolete Information

Due to 'goal-directed talk', which did not take place in the initial study, 'hallucinations' (i.e., misinformation about a restaurant, bus times, TV series, opera, and exaggerating robot capabilities) and obsolete information (e.g., weather, old movie recommendation) came into sight ($M = 0.57$, $SD = 0.84$) in 11 of the interactions, occurring up to 3 times. This led to neutral trust in the robot ($Md = 3.0$, $IQR = 0.25$), and caused varied perceptions of the correctness of the robot's responses ($Md = 3.5$, $IQR = 1.0$), and competency of the robot ($Md = 3.0$, $IQR = 1.0$). But it also led one participant to doubt themselves about the information they know (about the production date of a TV series)[22].

One of the participants noted in the interviews that they had tried asking the same questions (about the production date of a sailboat and whether they can sail tomorrow) that they had previously asked to ChatGPT, and the responses from the robot were "much worse", that the robot gave incorrect information about the weather, whereas ChatGPT told him, "I cannot forecast weather".

5.4.6 Disengagement Cues and Premature Closures

Disengagement cues from the robot, such as "I understand", and "That is good to know", brought the conversation to a halt, with participants not knowing how to respond, in 5 of the interactions, up to 4 times in an interaction ($M = 0.39$, $SD = 0.99$). Moreover, the robot tried to end the conversation prematurely (earlier than 7 minutes) in 7 interactions, up to 2 times ($M = 0.32$, $SD = 0.61$). The participant either changed the topic to continue the interaction without intervention, or the experimenter told them they could continue if they want to. This behavior might have led to mostly positive, but mixed opinions for the engagingness of the conversation ($Md = 3.0$, $IQR = 1.25$).

Topics were less frequently changed in comparison to the initial study, with several levels of follow-up questions in most interactions. However, in 3 of the interactions, the robot asked whether they would like to talk about another topic 3 or 4 times to the participants. In one of the interviews, a participant hypothesized that this behavior was due to the robot's lack of knowledge on the particular topic discussed.

6 Discussion

Minor technical improvements changed the user experience drastically. However, these developments were not sufficient to remove completely the presence of any of the challenges. This section discusses solutions to help address these challenges further.

6.1 Turn-taking

Both studies contrastively show the importance of turn-taking in HRI for user experiences, and that the disruptions in speech can have negative effects on older adults. While gaze aversion and light indication helped decrease the interruptions, but they still occurred in most of the conversations. Turn-taking behavior can be further

[22] https://youtu.be/8sHknc1mbhM

improved by using dialogue context (e.g., [75]), speech prosody (e.g., [76]), estimating the user's gaze, gestures, and facial expressions, such as eyebrow movement and mouth opening [77], or a combination of these features, e.g., [78–80] (for an in-depth review of turn-taking in HRI and conversational systems, see [81]) to prevent interrupting the user, both for improving user experience and speech recognition.

The long response generation time, which caused unnatural interactions, can be decreased by using a smaller LLM (with the trade-off of task performance or factuality [82]), an LLM with faster inference, quantization techniques [83], or by using conversation summaries rather than the full dialogue to decrease the prompt length. However, summaries may result in further hallucinations, which could be detrimental to the interaction as well as to the personalization (of learned facts).

To overcome awkward silences during the response generation as noted by the participants, fillers, such as "Hmm" and "Let me think", can be used [84]. However, overuse of them might also lead to the robot being perceived as repetitive. In addition, to decrease the superficialness of turns (sequential nature), backchannelling can be added using both verbal (e.g., "mm hm", "uh huh", "yeah") and non-verbal cues (e.g., nods, head shakes, facial expressions) [85], but it is important to do this in context with the dialogue, otherwise, it may also further deteriorate the user experience. Random gaze aversion should not be made during robot speech or while listening, as noted by two participants, because keeping eye contact with the user helps maintain the user engagement, perceived sociability, and improve quality of interaction [86].

6.2 Repetitive Responses

The increasing use of ChatGPT and other GPT-3.5 models, in addition to OpenAI prioritizing Plus subscribers in access to the servers, result in frequent response generation failures, as experienced in both studies, which can be overcome through multiple response requests. However, this increases the already long response time. Thus, using another LLM, such as LLaMA (Meta) [87] or its variants (e.g., Alpaca[23]), might be more suitable to achieve more reliable response generation. However, in the case of the response generation failures, the conversation can be pushed forward by invitation for elaboration, in addition to clarification requests, but they should be varied, and used to a minimum to avoid backlash from the user.

Large language models tend to be repetitive, and can get stuck in a loop [88], as evidenced in both studies, resulting in user frustration. The 'frequency penalty' parameter in GPT-3.5 can be increased to enforce the model to produce more varied responses (words). This can be further accompanied by sampling [82, 89], unlikelihood training [90], best-first decoding [91], or reinforcement learning from demonstration [92] to reduce repetitive behavior. To reduce the LLM from getting conditioned on itself to produce the same response again, prompt initializers can be used (e.g., "Rephrase the sentence") if the previously generated response is repeated outside of user clarification requests.

[23] https://github.com/tatsu-lab/stanford_alpaca

6.3 Superficial Conversations

To decrease the superficialness of conversations and provide a more natural and engaging interaction to older adults in their daily lives, the range of topics discussed can be more varied using the 'presence penalty' and 'temperature' in GPT-3.5. However, this may lead to more abrupt changes in topics, instead of smooth transitions with follow-ups (i.e., exploration vs exploitation trade-off). Fine-tuning can be applied with data on older adult speech to provide address this issue [19], however, that might also limit the model's goal-directed capabilities.

In addition, the persona prompt can be modified to use more follow-up questions and give deeper personality (e.g., pre-defined preferences, dislikes, memories) to the robot to improve the believability of the character, instead of starting from 'scratch'. The believability can further be improved by using mood and emotions for the agent, in addition to detecting the user's emotions to adapt the conversation accordingly [93]. Emotions can also be used as unsupervised RLHF signals for task performance [94], such that frustration in a conversation topic or phrase can be avoided in future interactions.

To generate deeper conversations that rely on shared history with the user, their preferences, dislikes, and daily activities, the robot should learn from the user. However, over long-term interactions, storing information in a database is not scalable, as prompts based on an ever-growing database would be too long. In addition, using a retrieval-based method and a database to update the results of LLMs with new data can result in hallucinations [95]. One option is to retrain models with new information, however, this would be computationally inefficient with a vast amounts of data, as well as result in 'catastrophic forgetting' of previously learned information [96, 97]. 'Lifelong (or continual) learning' [98] aims to address this problem by continually learning (the parameters of the model) over time to accommodate new knowledge (i.e., plasticity) while retaining previously learned information (i.e., stability). Within these methods, combining LLMs with 'parameter-expansion' methods offer the best balance of stability and plasticity [99]. Lifelong learning can help personalize the interactions, which in turn can help mitigate the negative user experiences that may arise due to poor performance in dialogue [100], improving user engagement [101], acceptability, and trust [102].

Over long-term deployments in real-world environments (e.g., older adults' homes or senior care centers), it is important to have full autonomy, such that the user can start the interaction without the need of a wizarded system. For a personalized companion robot, this requires detecting new users and incrementally learn them to recognize them to prevent sharing the information learned about the user with others, hence, an 'open world' user recognition algorithm, such as Multi-modal Incremental Bayesian Network [103], can be used. With personalization, the privacy of the interaction should be considered with utmost importance. Since current offline LLMs require vast computing power, it would not be possible to put them on a robot, but online models, as we did in our studies, risk the privacy of the individual. While the first interaction with a robot, as evidenced in our results, are mostly superficial, deeper conversations can be achieved over long-term interactions, in which users can talk about sensitive topics. The risks, trade-offs, and possible solutions to achieve privacy

are discussed in depth in the works by Bommasani et al. [104] and Weidinger et al. [105], which should be considered in parallel to the challenges highlighted in this work, while developing companion robots for older adults with LLMs.

6.4 Language Barrier

While speech recognition failures were low in the second study, their rare presence led to abrupt disruptions in the conversation flow. Since older adult speech contains pauses, and hesitancy, and is slower than younger adult speech, it is challenging to obtain good performance in speech recognition. One way to address this is to evaluate different speech recognition algorithms with older adult speech, such as Microsoft Azure or Whisper JAX[24], which optimizes Whisper for speed that can enable real-time transcription of the user utterances, and choose the algorithm that works the best.

Since GPT-3.5 allows changing the conversation language when the prompt is in a different language, enabling multi-lingual speech recognition would improve the capabilities of the robot, such as second language teaching as suggested by the participants in both studies, and help adapt to foreign visitors. However, allowing multiple languages to be transcribed may result in further interaction failures, especially due to overlapping words in different languages that may mean different things.

Also, the speech recognition output from Google Cloud does not contain punctuation marks, which may lead to disambiguity in speech. Parsing the responses may help improve the response generation from GPT-3.5.

6.5 Hallucinations and Obsolete Information

Disinformation can be risky in older adults' lives. For instance, if the user asks for advice on medicine, and the LLM produces a wrong answer, the consequences can be critical. Thus it is important to ensure that correct information is provided by LLMs. One approach is to prevent response generation in sensitive topics with filtering [105]. The hallucinations in the model can also be mitigated by applying inductive attention (e.g., [106]) or retrieval-based methods (e.g., [107]). Ji et al. [10] provide a detailed survey on hallucinations in LLMs with further suggestions.

In addition, due to their training data cut-off dates, LLMs contain obsolete information, which was present in the interactions in the second study, that can be addressed also by fact-checking in knowledge bases [108]. As suggested in Section 6.3, lifelong learning can be applied to update the information over time for user facts. However, for facts that can be obtained through the internet (such as weather), LLMs with browsing capabilities can be used (e.g., AutoGPT[25]).

6.6 Disengagement Cues and Premature Closures

Our work aims to design a personal companion robot that decreases loneliness in older adults, and supports them in their daily lives. If a robot tries to end a conversation in a short duration, this could further deepen their feelings of loneliness. Hence, it is important to engage the user in the conversation. One possible solution is to detect

[24] https://huggingface.co/spaces/sanchit-gandhi/whisper-jax
[25] https://github.com/Significant-Gravitas/Auto-GPT

engagement in the conversation in both user's and agent's responses, as well as through gaze, and other non-verbal or contextual information [101]. Upon detection of disengagement, a prompt can be used to change the topic, ask the user what they would like to talk about, or make the current topic more interesting, to encourage continuation in the conversation.

7 Limitations and Future Work

This work identified the challenges in applying LLMs to companion robots for open-domain dialogue with older adults as frequent interruptions and slow responses, repetitive responses, superficial conversations, language barrier, hallucinations and obsolete information, and disengagement cues. These challenges are by no means an exhaustive list, but aim to provide initial guidelines towards developing companion robots that can be part of the everyday lives of older adults. They also show the complexity of multi-modal interactions, in comparison to the text-based interactions that LLMs are typically evaluated on, especially for a population that is not familiar with the current state of the technology.

The LLM (GPT-3.5) and robot (Furhat) used in this work, in addition to the technical design decisions made, have led to the identification of dialogue disruptions that has and could negatively affect older adults' experiences with companion robots. However, the identified challenges may not replicate in other LLMs or robots, depending on their open-domain dialogue capabilities, modalities used, and the additional libraries that can address these challenges.

Moreover, as previously described, in both studies only a few participants (2 in the first study, 5 in the second) had previously interacted with a robot, with 2 participants talking to a wizarded robot prior to the first study, and only 1 participant talking to an (unknown) robot prior to the second study. Hence, this was the participants' ever first interaction with a (social) robot in the majority of the cases, leading to 'novelty effect' [109, 110] (i.e., users' perceptions and behaviors are affected by the novelty of the technology). Furthermore, talking to a robot as part of an experiment rather than having the robot "in the wild" (at home or a senior care center) may lead to changes in the behavior changes due to being observed [111] and create an artificial context within conversations, preventing to unlock the full range of topics in open-domain conversation and corresponding challenges that can arise in conversations with a companion robot over long-term interactions [112]. In addition, the participants in both studies had volunteered (with only a small compensation offered at the second study), which indicates an interest in robots (i.e., 'participation bias') [111] and may not reflect the views of the general population of older adults. Nonetheless, the aim of these studies were to identify the primary obstacles that may arise in conversations and overcome them, in addition to understanding needs, preferences, and expectations of older adults through iterative co-design, prior to exposing them to new technology in their homes or senior care centers to avoid backlash with fear, annoyance, or reluctance to use robots.

In future work, we intend to develop the robot's capabilities further using the solutions suggested in this section, and continue the iterative co-design with older

adults, by bringing back the participants from both studies for another interaction with the robot. After confirming that the robot is suitable for deployments, we intend to deploy it for a real-world long-term study in older adults' homes, which can overcome the unnaturalness of conversations and perhaps show deeper challenges to address.

While the challenges and the corresponding suggestions presented here for applying LLMs to companion robots are based on the interactions with older adults, they may hold for socially assistive robots or companion robots with other populations, as well as general-purpose robots and spoken dialogue systems, in other words, wherever an open-domain conversation may take place.

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Declarations

Compliance with Ethical Standards

The study was conducted according to the standards from Ethical Review Authority in Sweden. All participants gave an informed consent to participate in the studies. Participants, whose images and/or videos appear in this article, gave informed consent for anonymized (i.e., blurred face, and without full name released) image and/or video sharing in publications (conferences and/or journals). The study did not collect any sensitive or health-related information from the participants.

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Conflicts of Interest

G.S. is co-affiliated with Furhat Robotics, as its Co-founder and Chief Scientist. The remaining authors have no relevant financial or non-financial interests to disclose.

Usage of a Large Language Model in the Article

B.I. used ChatGPT (OpenAI) to generate ideas for the paper title, section headers, terminology, and synonyms for words. The information generated by the model was never used directly: it was modified, fact-checked, and combined with the author's own ideas and text.

Table A1: Hyperparameters of GPT-3.5 used in both studies. The default values from the OpenAI Playground for Chat preset were used, except for the maximum length and frequency penalty (see footnotes).

Hyperparameter	Value	Range
Temperature	0.9	[0,1]
Maximum Length	50[1]	[1,4000]
Top P	1.0	[0,1]
Frequency Penalty	0.5[2]	[0,2]
Presence Penalty	0.6	[0,2]
Stop Words	*USER_NAME*:, Furhat:	
Best of	1	[1,20]

[1]The maximum length for tokens was decreased (default value 150) to generate shorter responses to keep the conversation flowing and engaging the user.

[2]The frequency penalty was increased (default value 0) to decrease the model's likelihood to repeat the same responses. Because the users may ask the robot to repeat its responses (e.g., when they don't hear it well), this value was only increased to 0.5.

Appendix A Hyperparameters of GPT-3.5

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